
3rd International Conference on Networks, Multimedia, and Information Technology (NMITCON-2025) : Submission (2061) has been created.

From Microsoft CMT <noreply@msr-cmt.org>

Date Sun 4/27/2025 5:47 PM

To SHARMA SWASTIK <1022251@COMP.fcrit.ac.in>

Hello,

The following submission has been created.

Track Name: NMITCON2025

Paper ID: 2061

Paper Title: SmartChain: Elevating Supply Chain Efficiency with AI and Blockchain

Abstract:

Traditional supply chain management systems are hindered by persistent challenges, including widespread fraud, operational inefficiencies, and a critical lack of transparency, all of which contribute to substantial financial losses and delays in goods delivery. To overcome these limitations, this paper introduces SmartChain, an innovative framework that seamlessly integrates blockchain technology with artificial intelligence (AI) to establish a robust, secure, and highly efficient supply chain ecosystem. SmartChain leverages Ethereum-based smart contracts to automate and secure critical supply chain processes, such as procurement and logistics, ensuring that all movements of goods are recorded in an immutable and transparent ledger accessible to authorized stakeholders. Simultaneously, advanced AI models enable precise demand forecasting, identify patterns indicative of fraudulent behavior, and optimize inventory management, reducing waste and enhancing operational resilience. The design of SmartChain emphasizes scalability and adaptability, making it suitable for diverse industries, from manufacturing to retail. This paper provides a detailed exploration of SmartChain's architecture, including its implementation using real-world-inspired simulations and its evaluation across multiple performance metrics. Results from these simulations reveal that SmartChain achieves a transaction accuracy rate of 99.8%, improves fraud detection by 30% compared to traditional methods, and boosts inventory efficiency by 20%, demonstrating its superiority over conventional systems. By addressing the core deficiencies of traditional supply chains, SmartChain enhances trust, reduces costs, and streamlines operations through enhanced transparency in the transfer of goods. This work contributes to the growing field of blockchain-AI integration, providing a practical blueprint for modernizing supply chain management with cutting-edge technology.

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Authors:

- 1022251@comp.fcrit.ac.in (Primary)
- 1022219@comp.fcrit.ac.in
- 1022226@comp.fcrit.ac.in
- 1022247@comp.fcrit.ac.in

Secondary Subject Areas: Not Entered

Submission Files:

SmartChain_Paper (5).pdf (1007 Kb, Sun, 27 Apr 2025 12:16:54 GMT)

Submission Questions Response: Not Entered

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